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IMAGE PROCESSING APPARATUS, IMAGE PROCESSING METHOD, AND STORAGE MEDIUM

Abstract

Three-dimensional information which enables generation of a high-quality virtual viewpoint image is estimated while reducing the amount of computations for estimating the three-dimensional information. An image processing apparatus according to this disclosure obtains data on a captured image, extracts a high-frequency pattern region from the captured image, generates a plurality of ray groups based on rays corresponding to each of a plurality of pixels included in the captured image, and learns three-dimensional information on a space using the plurality of ray groups, wherein the image processing apparatus generates the plurality of ray groups including one or more high-frequency pattern region ray groups which are ray groups where a density of rays corresponding to pixels included in the high-frequency pattern region is higher than a density of rays corresponding to pixels included in a region other than the high-frequency pattern region.

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Background/Summary

BACKGROUND

Field

[0001] The present disclosure relates to a technique to estimate three-dimensional information about a space including an object.

Description of the Related Art

[0002] There is a technique of estimating information (hereinafter referred to as “three-dimensional information”) about a space including an object using images (hereinafter referred to as “captured images”) obtained by capturing the object from various directions. There is also a technique of generating an image (hereinafter referred to as “virtual viewpoint image”) corresponding to a representation of an object viewed from an arbitrary virtual viewpoint (hereinafter referred to as “virtual viewpoint”) using three-dimensional information. However, in a case where the object has a complicated three-dimensional shape, color, reflection property, or the like, there is a possibility that three-dimensional information about a space including the object cannot be correctly estimated, which may result in deterioration in image quality of the virtual viewpoint image.

[0003] Japanese Patent Laid-Open No. 2023-66705 (hereinafter referred to as “Patent Literature 1”) discloses a technique of learning, as three-dimensional information, radiance fields which represent colors and densities according to positions and directions in a space including an object using captured images as training images. Patent Literature 1 also discloses a technique of generating a virtual viewpoint image by volume rendering using the radiance fields estimated by learning. Specifically, in the technique disclosed in Patent Literature 1, learning parameters corresponding to the radiance fields are calculated by sampling and machine learning of a point on a ray corresponding to each pixel of a training image. More specifically, in the technique disclosed in Patent Literature 1, in the calculation, a sampling density of a ray corresponding to each pixel of the training image within the depth of field is higher than a sampling density out of the depth of field. The technique disclosed in Patent Literature 1 controls the sampling density based on the depth of field, thereby improving the estimation accuracy of radiance fields of a space corresponding to an object within the depth of field while reducing the amount of computations for estimating the radiance fields and improving the image quality of the virtual viewpoint image.

SUMMARY

[0004] However, the inventor(s) herein recognized that since the technique disclosed in Patent Literature 1 uniformly reduces the sampling density of rays out of the depth of field, the radiance fields of a space corresponding to an object out of the depth of field cannot be estimated with high accuracy.

[0005] Thus, the present disclosure aims to provide a technique to estimate three-dimensional information which enables generation of a high-quality virtual viewpoint image while reducing the amount of computations for estimating the three-dimensional information.

[0006] An image processing apparatus according to the present disclosure comprises: one or more hardware processors; and one or more memories storing one or more programs configured to be executed by the one or more hardware processors, the one or more programs including instructions for: obtaining data on a captured image; extracting a high-frequency pattern region from the captured image; generating a plurality of ray groups based on rays corresponding to respective pixels included in the captured image; generating the ray groups including one or more high-frequency pattern region ray groups which are ray groups where a density of rays corresponding to pixels included in the high-frequency pattern region is higher than a density of rays corresponding

to pixels included in a region other than the high-frequency pattern region; and learning three-dimensional information on a space using the ray groups.

[0007] Further features of various embodiments will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a diagram showing an example of a configuration of an image processing system according to a first embodiment;

[0009] FIG. 2 is a block diagram showing an example of a hardware configuration of an image processing apparatus according to the first embodiment;

[0010] FIG. 3 is a block diagram showing an example of a functional configuration of the image processing apparatus according to the first embodiment;

[0011] FIG. 4 is a flowchart showing an example of a processing flow in the image processing apparatus according to the first embodiment;

[0012] FIG. 5A is a diagram showing an example of arrangement of image capturing apparatuses according to the first embodiment and FIGS. 5B to 5D are diagrams showing an example of captured images;

[0013] FIGS. 6A to 6C are diagrams for illustrating an extraction process of a high-frequency pattern region in an extraction unit according to the first embodiment;

[0014] FIG. 7 is a flowchart showing an example of the flow of a generation process of ray groups in a ray group generation unit 303 according to the first embodiment;

[0015] FIGS. 8A and 8B are diagrams for illustrating examples of ray groups generated by the ray group generation unit according to the first embodiment;

[0016] FIGS. 9A and 9B are diagrams for illustrating an example of a ray corresponding to each piece of ray information included in ray groups according to the first embodiment;

[0017] FIG. 10 is a flowchart showing an example of the flow of a learning process of three-dimensional information in a learning unit according to the first embodiment;

[0018] FIG. 11 is a flowchart showing an example of the flow of the extraction process of the high-frequency pattern region in the extraction unit according to a second embodiment;

[0019] FIGS. 12A to 12D are diagrams for illustrating the extraction process of the high-frequency pattern region in the extraction unit according to the second embodiment;

[0020] FIG. 13 is a diagram showing an example of an approximate shape of an object obtained by a visual hull method;

[0021] FIG. 14 is a flowchart showing an example of the flow of the generation process of ray groups in the ray group generation unit according to the second embodiment;

[0022] FIGS. 15A and 15B are diagrams for illustrating examples of ray groups generated by the ray group generation unit according to the second embodiment; and

[0023] FIGS. 16A and 16B are diagrams for illustrating an example of a ray corresponding to each piece of ray information included in ray groups according to the second embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0024] Hereinafter, with reference to the attached drawings, the present disclosure is explained in detail in accordance with preferred embodiments. Configurations shown in the following embodiments are merely exemplary and the present disclosure is not limited to the configurations shown schematically.

First Embodiment

[0025] In the present embodiment described below, radiance fields corresponding to a space including an object are learned based on data on captured images (hereinafter referred to as

“captured image data”) obtained by capturing the object with a plurality of image capturing apparatuses from various directions. Specifically, the above learning in the present embodiment uses ray groups selected based on an image region (hereinafter referred to as “high-frequency pattern region”) including a representation of a high-frequency pattern extracted from each captured image so that the density of rays is high in a space including the high-frequency pattern.

<Configuration of Image Processing System>

[0026] FIG. 1 is a diagram showing an example of a configuration of an image processing system according to the first embodiment. The image processing system comprises a plurality of image capturing apparatuses **101**, an image processing apparatus **102**, a user interface (hereinafter referred to as “UI”) panel **103**, a storage apparatus **104**, and a display apparatus **105**. The plurality of image capturing apparatuses **101** are formed by digital still cameras, digital video cameras, or the like and arranged in different positions each other. The image capturing apparatuses **101** each capture an object **107** existing in an image capturing region **106** from various directions in synchronization with each other according to a predetermined image capturing condition and output captured image data obtained through the image capturing to the image processing apparatus **102**. The captured image data obtained through image capturing by the image capturing apparatuses **101** may be either or both of still image data and moving image data. The term “image” used herein means both “still image” and “moving image” unless otherwise specified.

[0027] The image processing apparatus **102** obtains data on captured images output from the plurality of image capturing apparatuses **101** and learns information (three-dimensional information) about a three-dimensional shape and color of a space including the object **107** existing in the image capturing region **106** generated using the obtained captured images. Further, the image processing apparatus **102** generates a virtual viewpoint image based on the three-dimensional information obtained as a result of learning. Incidentally, the three-dimensional information in the image capturing region **106** to be learned differs according to the content of learning. In the first embodiment, a radiance field is assumed as an example of the three-dimensional information to be learned.

[0028] Although the present embodiment is based on the assumption that each of the plurality of image capturing apparatuses **101** is connected to the image processing apparatus **102** as shown in FIG. 1, the method of connection between the image capturing apparatuses **101** and the image processing apparatus **102** is not limited to this. Specifically, for example, the plurality of image capturing apparatuses **101** may be cascaded by connecting the image capturing apparatuses **101** adjacent to each other and at least one of the plurality of image capturing apparatuses **101** may be connected to the image processing apparatus **102**.

[0029] Further, although the present embodiment is based on the assumption that the plurality of image capturing apparatuses **101** are arranged in different positions each other, the number and arrangement of image capturing apparatuses **101** are not limited to this. For example, in a case where the position, shape, and color of the object **107** existing in the image capturing region **106**, the intensity or color of environmental light, and the like are not changed with time, at least one image capturing apparatus **101** whose position and orientation can be changed may be arranged. In this case, the image capturing apparatus **101** may capture images in different positions while changing its position and orientation and the image processing apparatus **102** may obtain data on the plurality of captured images obtained through the image capturing.

[0030] The UI panel **103** comprises a display device such as a liquid-crystal panel and displays on the display device a user interface to provide a user with information such as image capturing conditions in the image capturing apparatuses **101** and processing settings of the image processing apparatus **102**. The UI panel **103** may also comprise an input device such as a touch panel or button. In this case, the UI panel **103** accepts an instruction from a user about a change in the above image capturing conditions or processing settings, or the like. The input device may be provided separately from the UI panel **103** as a mouse, keyboard, or the like.

[0031] The storage apparatus **104** is formed by a hard disk drive or the like and stores data on a virtual viewpoint image output from the image processing apparatus **102**. In a case where the image processing apparatus **102** outputs three-dimensional information, the storage apparatus **104** may store the three-dimensional information output from the image processing apparatus **102**. The display apparatus **105** is formed by a liquid-crystal display or the like, obtains an image signal indicating a virtual viewpoint image output from the image processing apparatus **102**, and displays the virtual viewpoint image corresponding to the image signal. In a case where the image processing apparatus **102** outputs an image signal indicating three-dimensional information, the display apparatus **105** may obtain an image signal indicating three-dimensional information output from the image processing apparatus **102** and display an image corresponding to the image signal. The image capturing region **106** is a three-dimensional region surrounded by the plurality of image capturing apparatuses **101** installed in a studio or the like. The frame shown by the solid line in FIG. **1** represents the outline of the image capturing region **106** on the floor.

<Hardware Configuration of Image Processing Apparatus>

[0032] FIG. **2** is a block diagram showing an example of a hardware configuration of the image processing apparatus **102** according to the first embodiment. As the hardware configuration, the image processing apparatus **102** comprises a CPU **201**, a RAM **202**, a ROM **203**, a storage device **204**, a control interface (hereinafter referred to as “I/F”) **205**, an input I/F **206**, an output I/F **207**, and a main bus **208**. The CPU **201** is a processor which has control over each unit of the image processing apparatus **102**. The RAM **202** functions as main memory and work area of the CPU **201** and the like. The ROM **203** stores a group of programs executed by the CPU **201**. The storage device **204** is formed by a hard disk drive or the like and stores an application program executed by the CPU **201**, data used for processing of the CPU **201**, and the like.

[0033] The control I/F **205** is a communication interface which is connected to each image capturing apparatus **101** and is used to control settings of image capturing conditions for each image capturing apparatus **101**, start image capturing, stop image capturing, and the like. The input I/F **206** is a communication interface using a serial bus or the like such as a Serial Digital Interface (SDI) or High-Definition Multimedia Interface (HDMI) (registered trademark). Captured image data output from each image capturing apparatus **101** is obtained via the input I/F **206**. The output I/F **207** is a communication interface using a serial bus or the like such as a Universal Serial Bus (USB) or DisplayPort (registered trademark). Data or an image signal of a virtual viewpoint image or the like is output to the storage apparatus **104** or the display apparatus **105** via the output I/F **207**. The main bus **208** is a transmission line which connects the above hardware elements of the image processing apparatus **102** so that they can communicate with each other.

<Functional Configuration of Image Processing Apparatus>

[0034] FIG. **3** is a block diagram showing an example of a functional configuration of the image processing apparatus **102** according to the first embodiment. The image processing apparatus **102** comprises an image obtaining unit **301**, an extraction unit **302**, a ray group generation unit **303**, a learning unit **304**, a viewpoint obtaining unit **305**, an image generation unit **306**, and an output unit **307**. Each unit the image processing apparatus **102** comprises as the functional configuration is implemented by the CPU **201** executing a program stored in the ROM **203** using the RAM **202** as work memory. Incidentally, not all of the following processes of the units the image processing apparatus **102** comprises as the functional configuration have to be executed by the CPU **201**; part or all of the processes may be executed by one or more processing circuits other than the CPU **201**.

[0035] The image obtaining unit **301** obtains captured image data obtained by each image capturing apparatus **101** capturing the image capturing region **106** and parameters of image capturing (hereinafter referred to as “image capturing parameters”) in the image capturing apparatuses **101** corresponding to the captured images. The extraction unit **302** extracts a high-frequency pattern region from the captured images obtained by the image obtaining unit **301**. The extraction process of the high-frequency pattern region in the extraction unit **302** will be described later in detail.

Based on the captured images and image capturing parameters obtained by the image obtaining unit **301** and the high-frequency pattern region extracted by the extraction unit **302**, the ray group generation unit **303** generates a plurality of ray groups including such a ray group that the density of rays is high in a space including the high-frequency pattern. The generation process of ray groups in the ray group generation unit **303** will be described later in detail.

[0036] The learning unit **304** learns information (three-dimensional information) about radiance fields of a space including the object **107** based on the ray groups generated by the ray group generation unit **303**. The three-dimensional information is, for example, a network parameter in a learning model formed by a multilayer perceptron (MLP) representing the radiance fields of the space including the object **107**. The learning process of the three-dimensional information in the learning unit **304** will be described later in detail.

[0037] The viewpoint obtaining unit **305** obtains information about a virtual viewpoint (hereinafter referred to as “virtual viewpoint information”). The virtual viewpoint information is information indicating a position of a virtual viewpoint and a line-of-sight direction at the virtual viewpoint and corresponding to an image capturing parameter (hereinafter referred to as “virtual camera parameter”) of a virtual image capturing apparatus (hereinafter referred to as “virtual camera”) arranged at the virtual viewpoint. The image generation unit **306** generates a virtual viewpoint image using a learned model which is the three-dimensional information obtained as a result of learning by the learning unit **304**, that is, information about radiance fields, and the virtual camera parameter obtained by the viewpoint obtaining unit **305**. Specifically, the image generation unit **306** performs volume rendering using the three-dimensional information and thereby generates a virtual viewpoint image corresponding to the virtual viewpoint indicated by the virtual camera parameter.

[0038] The output unit **307** outputs data on the virtual viewpoint image generated by the image generation unit **306** to the storage apparatus **104** and causes the storage apparatus **104** to store the data. The output unit **307** may output the virtual viewpoint image as an image signal to the display apparatus **105** and cause the display apparatus **105** to display the virtual viewpoint image. The output unit **307** may also output the learned model, which is the three-dimensional information obtained as a result of learning by the learning unit **304**, that is, information about radiance fields, to the storage apparatus **104** or the like.

<Operation of Image Processing Apparatus>

[0039] FIG. **4** is a flowchart showing an example of a processing flow in the image processing apparatus **102** according to the first embodiment. In the following description, each processing step is denoted by a reference numeral headed with “S.” In a case where each image capturing apparatus **101** outputs moving image data as the captured image data, the image processing apparatus **102** repeats the process of this flowchart each time data on a frame of the moving image obtained through synchronized image capturing is output from the image capturing apparatus **101**. First, in **S401**, the image obtaining unit **301** obtains data on captured images obtained through the image capturing and an image capturing parameter for the image capturing apparatus **101** corresponding to each captured image. Specifically, for example, the captured image data is obtained from the image capturing apparatus **101** via the input I/F **206**. The image capturing parameter is calculated by execution of calibration or the like and stored in the storage device **204** in advance, and is then obtained by being read. The captured image data and image capturing parameters obtained in **S401** are stored in the RAM **202**.

[0040] FIG. **5A** is a diagram showing an example of arrangement of the image capturing apparatuses **101** according to the first embodiment and FIGS. **5B** to **5D** are diagrams showing an example of captured images **501** to **503** obtained through image capturing by image capturing apparatuses **101a** to **101c**. FIG. **5A** shows an example of arrangement of the image capturing apparatuses **101**, where the image capturing apparatuses **101** are arranged so that the object **107** existing in the image capturing region **106** can be captured from various directions. Incidentally, the image capturing apparatuses **101a**, **101b**, and **101c** shown in FIG. **5A** are identical to the other

image capturing apparatuses **101**. FIG. 5B shows an example of the captured image **501** obtained through image capturing by the image capturing apparatus **101a**, FIG. 5C shows an example of the captured image **502** obtained through image capturing by the image capturing apparatus **101b**, and FIG. 5D shows an example of the captured image **503** obtained through image capturing by the image capturing apparatus **101c**.

[0041] After **S401**, in **S402**, the extraction unit **302** executes an extraction process of a high-frequency pattern region for each captured image obtained in **S401** and thereby extracts the high-frequency pattern region from each captured image obtained in **S401**. For example, the extraction unit **302** uses a differential filter to specify an image region in which the density of edges of a striped pattern or the like is higher than a predetermined density from each captured image and extracts the specified image region as the high-frequency pattern region. Specifically, for example, the extraction unit **302** first applies a Laplacian filter to the captured image to extract a plurality of pixels with an absolute value of filter output greater than a predetermined threshold as an edge region. Next, the extraction unit **302** sequentially applies, to the extracted edge region, hole removing processing by closing processing and small object removing processing by opening processing, and extracts the result thus obtained as the high-frequency pattern region.

[0042] FIGS. **6A** to **6C** are diagrams for explaining the extraction process of the high-frequency pattern region in the extraction unit **302** according to the first embodiment. FIG. **6A** shows an example of a captured image **600** obtained through image capturing by an image capturing apparatus **101**. For example, it is assumed that the captured image **600** shown in FIG. **6A** includes representations **601** to **603** of three objects existing in the image capturing region **106**. FIG. **6B** is a diagram showing an example of an image (hereinafter referred to as “edge image”) **610** indicating edge regions **611** extracted from the captured image **600** by the extraction unit **302**. In the edge image **610** illustrated in FIG. **6B**, pixels corresponding to the edge regions **611** are shown in white and pixels corresponding to a region other than the edge regions **611** are shown in black. FIG. **6C** shows an example of an image (hereinafter referred to as “high-frequency pattern image”) **620** indicating a high-frequency pattern region **621** extracted from the edge image **610** by the extraction unit **302**. In the high-frequency pattern image **620** illustrated in FIG. **6C**, pixels corresponding to the high-frequency pattern region **621** are shown in white and pixels corresponding to a region other than the high-frequency pattern region **621** are shown in black.

[0043] After **S402**, in **S403**, the ray group generation unit **303** executes a generation process of ray groups and generates a plurality of ray groups based on the captured images and image capturing parameters obtained in **S401** and the high-frequency pattern region extracted in **S402**. Specifically, the ray group generation unit **303** generates a plurality of ray groups including such a ray group that the density of rays is high in the space including the high-frequency pattern. The generation process of ray groups in the ray group generation unit **303** will be described later in detail. Next, in **S404**, the learning unit **304** executes a learning process of three-dimensional information and learns a learning model indicating three-dimensional information on the space including the objects, that is, radiance fields in the space, based on the ray groups generated in **S403**. The learning process of three-dimensional information in the learning unit **304** will be described later in detail.

[0044] Next, in **S405**, the learning unit **304** judges whether to finish the learning process of three-dimensional information. Specifically, the learning unit **304** compares the captured images obtained in **S401** with images generated by volume rendering based on the image capturing parameters corresponding to the captured images and the three-dimensional information (learning model representing radiance fields) learned in **S404**. As a result of the comparison, in a case where a difference between the captured images and the images generated by volume rendering is less than a predetermined threshold, the learning unit **304** judges that the learning process of three-dimensional information is to be finished. In a case where the difference is greater than the predetermined threshold, the learning unit **304** judges that the learning process of three-dimensional information is not to be finished. In a case where the learning unit **304** judges in **S405**

that the learning process of three-dimensional information is not to be finished, the image processing apparatus **102** returns to the process of **S403** and repeats the process from **S403** to **S405** until the learning unit **304** judges in **S405** that the learning process of three-dimensional information is to be finished.

[0045] In a case where the learning unit **304** judges in **S405** that the learning process of three-dimensional information is to be finished, the viewpoint obtaining unit **305** obtains in **S406** a virtual camera parameter set based on an instruction from a user using the UI panel **103**. The method of obtaining the virtual camera parameter by the viewpoint obtaining unit **305** is not limited to the above method. For example, the viewpoint obtaining unit **305** may obtain the virtual camera parameter by reading the virtual camera parameter which is set and stored in the storage device **204** in advance.

[0046] After **S406**, in **S407**, the image generation unit **306** generates a virtual viewpoint image using the virtual camera parameter obtained in **S406** and the three-dimensional information (learned model representing radiance fields) obtained as a result of the learning process in **S404**. Specifically, the image generation unit **306** generates a virtual viewpoint image corresponding to the virtual viewpoint indicated by the virtual camera parameter by performing volume rendering of the three-dimensional information (learned model representing radiance fields) based on the virtual camera parameter.

[0047] Next, in **S408**, the output unit **307** outputs the virtual viewpoint image generated in **S407**. Specifically, for example, the output unit **307** outputs data on the virtual viewpoint image or an image signal indicating the virtual viewpoint image to the storage apparatus **104**, the display apparatus **105**, or the like via the output I/F **207**. After **S408**, the image processing apparatus **102** finishes the process of the flowchart shown in FIG. **4**. Incidentally, as stated above, in a case where each image capturing apparatus **101** outputs moving image data as the captured image data, the image processing apparatus **102** returns to **S401** after **S408** and repeats the process of the flowchart.

<Generation Process of Ray Groups>

[0048] FIG. **7** is a flowchart showing an example of the flow of the generation process of ray groups in the ray group generation unit **303** according to the first embodiment and showing an example of the detailed processing flow in **S403** shown in FIG. **4**. In **S403**, a plurality of ray groups including such a ray group that the density of rays is high in the space including the high-frequency pattern are generated based on the captured images and image capturing parameters obtained in **S401** and the high-frequency pattern region extracted in **S402**. The process of this flowchart is executed after the process in **S402**.

[0049] After **S402**, in **S701**, the ray group generation unit **303** first obtains information (hereinafter referred to as “ray information”) about a ray corresponding to each pixel of the captured images based on the captured images and image capturing parameters obtained in **S401**. It is assumed in the present embodiment that the ray information includes values concerning a color, start point, and direction of the ray and the ray group generation unit **303** obtains the ray information by calculating these values. The ray group generation unit **303** obtains a value concerning the color of the ray on the assumption that the color of the ray is a color corresponding to a value of the corresponding pixel. The ray group generation unit **303** obtains a value concerning the start point of the ray on the assumption that the start point of the ray is a position of the corresponding image capturing apparatus **101**. For the direction of the ray, the ray group generation unit **303** obtains a value concerning the direction of the ray based on the image capturing parameter corresponding to the captured image and coordinates (hereinafter referred to as “image coordinates”) of a pixel in the captured image, for example, by performing a computation using the following equation (1):

[00001]
$$d = ((u - c_x) / f_x, (v - c_y) / f_y, 1) \quad \text{equation(1)}$$

[0050] Here, d is a vector representing a direction of a ray corresponding to image coordinates in a

case where two axes orthogonal to each other in the image are X and Y axes and an axis orthogonal to the X and Y axes is a Z axis. (u, v) are image coordinates and (c.sub.x, c.sub.y) are image coordinates corresponding to a center position of the image. f.sub.x is a value obtained by multiplying the number of light-sensitive elements per unit length in the X-axis direction of an image sensor of the image capturing apparatus **101** by a value indicating a focal length of an optical system of the image capturing apparatus **101**. Similarly, f.sub.y is a value obtained by multiplying the number of light-sensitive elements per unit length in the Y-axis direction of the image sensor of the image capturing apparatus **101** by the value indicating the focal length of the optical system of the image capturing apparatus **101**. (f.sub.x, f.sub.y) is also referred to as a focal length as an intrinsic parameter of the image capturing apparatus **101**. It is assumed here that the intrinsic parameter is included in the image capturing parameter.

[0051] After **S701**, in **S702**, the ray group generation unit **303** generates a plurality of ray groups based on ray information corresponding to image coordinates of each pixel included in the high-frequency pattern region (hereinafter simply referred to as “ray information corresponding to the high-frequency pattern region”). One ray group includes a plurality of pieces of ray information. In the present embodiment, it is assumed that a plurality of ray groups are generated so that the same piece of ray information corresponding to the high-frequency pattern region is not included in more than one ray group. Specifically, the ray group generation unit **303** defines a predetermined number of pieces of ray information selected randomly from the ray information corresponding to the high-frequency pattern region as one ray group. More specifically, the ray group generation unit **303** generates a plurality of ray groups so that the same piece of ray information is not selected multiple times from a plurality of pieces of ray information corresponding to the high-frequency pattern region. The ray group generation unit **303** repeats the above process until the number of unselected pieces of ray information corresponding to the high-frequency pattern region is less than a predetermined number. In this manner, a plurality of ray groups are generated so that the same piece of ray information corresponding to the high-frequency pattern region is not included in more than one ray group. Incidentally, unselected pieces of ray information corresponding to the high-frequency pattern region may be used as residual ray information in the process of **S703** to be described later.

[0052] FIGS. **8A** and **8B** are diagrams for explaining examples of ray groups generated by the ray group generation unit **303**. Specifically, FIG. **8A** is a diagram for explaining an example of ray groups according to the first embodiment and FIG. **8B** is a diagram for explaining an example of ray groups according to modified example 1 of the first embodiment to be described later. In FIGS. **8A** and **8B**, rectangles **841** filled in gray represent ray information corresponding to the high-frequency pattern region. Rectangles **842** filled in white represent ray information (hereinafter simply referred to as “residual ray information”) corresponding to image coordinates of each pixel included in a region (hereinafter referred to as “residual region”) other than the high-frequency pattern region. A set of ray information **851** represents a set of all pieces of ray information corresponding to the high-frequency pattern region, which is a set of pieces of ray information that are selectable in a case where the ray group generation unit **303** generates a plurality of ray groups in the process of **S702**. A set of ray information **852** represents a set of all the residual ray information. A set of ray information **850** represents a set of pieces of ray information corresponding to image coordinates of the all pixels of the captured image, including all the pieces of ray information corresponding to the high-frequency pattern region and all the residual ray information.

[0053] The ray group generation unit **303** executes the process of **S702** and thereby generates a plurality of ray groups **801** to **803** illustrated as an example in FIG. **8A**, each including only the ray information corresponding to the high-frequency pattern region. A ray corresponding to each piece of ray information included in the ray groups **801** to **803** thus generated intersects with the object including the high-frequency pattern.

[0054] FIGS. 9A and 9B are diagrams for explaining an example of a ray 904 corresponding to each piece of ray information included in the ray groups 801 to 803 shown in FIG. 8A. Specifically, FIG. 9A is a diagram of the image capturing region 106 viewed vertically from above and shows an example of the ray 904 corresponding to each piece of ray information included in the ray groups 801 to 803 shown in FIG. 8A. In FIG. 9A, an object 901 is an object including the high-frequency pattern and objects 902 and 903 are objects not including the high-frequency pattern. FIG. 9B shows an example of a captured image 910 obtained through image capturing by the image capturing apparatus 101c. The captured image 910 shown in FIG. 9B includes representations 911 to 913 of the three objects 901 to 903 existing in the image capturing region 106.

[0055] As described above, the ray 904 corresponding to each piece of ray information included in the ray groups 801 to 803 intersects with the object 903 including the high-frequency pattern. Accordingly, as shown in FIG. 9A, the density of rays passing through the space including the high-frequency pattern is higher than the density of rays passing through the other space. Thus, learning can be performed using high-density ray groups for the space corresponding to the object 901 including the high-frequency pattern. As a result, the three-dimensional information (radiance fields) on the space corresponding to the object 901 including the high-frequency pattern can be estimated with high accuracy.

[0056] After S702, in S703, the ray group generation unit 303 generates a plurality of ray groups based on the residual ray information. In the present embodiment, a plurality of ray groups are generated so that the same ray information in residual ray information is not included in more than one ray group. Specifically, a predetermined number of pieces of ray information selected randomly from the residual ray information are defined as one ray group. More specifically, the ray group generation unit 303 generates a plurality of ray groups so that the same ray information in residual ray information is not selected multiple times. The ray group generation unit 303 repeats the above process until the number of unselected pieces of residual ray information is less than a predetermined number. In this manner, a plurality of ray groups are generated so that the same ray information in residual ray information is not included in more than one ray group. Incidentally, ray groups 804 to 812 shown in FIG. 8A are an example of ray groups including only the residual ray information, which are generated by the ray group generation unit 303 executing the process of S703.

[0057] After S703, the ray group generation unit 303 finishes the process of the flowchart shown in FIG. 7, namely the process of S403 shown in FIG. 4. That is, the ray group generation unit 303 executes the process of S403 and thereby generates a plurality of ray groups 801 to 812 from the set of ray information 850 corresponding to the image coordinates of all the pixels included in the captured image, as illustrated as an example in FIG. 8A.

<Learning Process of Three-Dimensional Information>

[0058] FIG. 10 is a flowchart showing an example of the flow of the learning process of three-dimensional information in the learning unit 304 according to the first embodiment and showing an example of the detailed processing flow in S404 shown in FIG. 4. In S404, rendering values of rays are calculated for each ray group by volume rendering based on the plurality of ray groups generated in S403 and the three-dimensional information is updated so that a difference between the rendering values of the rays and values indicating colors of the rays becomes small. In the present embodiment, it is assumed that the three-dimensional information is a function (learning model) formed by a multilayer perceptron and indicating radiance fields, which function receives the input of information indicating a position and a vector obtained by encoding a direction and outputs a value indicating the corresponding density and color. The process of this flowchart is executed after the process of S403.

[0059] After S403, in S1001, the learning unit 304 first selects an arbitrary ray group from the ray groups generated in S403. Specifically, the learning unit 304 randomly selects an arbitrary one of unselected ray groups. The ray group selected in S1001 is hereinafter referred to as “ray group of

interest.” Next, in **S1002**, the learning unit **304** performs volume rendering using a plurality of pieces of ray information included in the ray group of interest and thereby calculates a rendering value of a ray corresponding to each piece of ray information. Specifically, in **S1002**, the learning unit **304** first sets a plurality of sampling points on a ray corresponding to each piece of ray information included in the ray group of interest based on a value concerning a start point of the ray and a value concerning a direction of the ray included in the piece of ray information. Next, in **S1002**, the learning unit **304** obtains a density and color corresponding to the positions of the sampling points and the direction of the ray based on the learning model indicating radiance fields and calculates the rendering value of the ray corresponding to the piece of ray information by, for example, performing a computation using the following equations (2) and (3):

$$[00002] \ C(r) = \text{Math}.\prod_{i=1}^N T_i(1 - \exp(-\sigma_{\text{sub}.i} \delta_{\text{sub}.i}))c_i \quad \text{equation(2)}$$

$$T_i = \exp(-\text{Math}.\prod_{i=1}^{i-1} \sigma_{\text{sub}.i} \delta_{\text{sub}.i}) \quad \text{equation(3)}$$

[0060] Here, $C(r)$ is a rendering value of a ray r , i and j are indices of sampling points, and N is a total number of sampling points. $\sigma_{\text{sub}.i}$ is a value indicating a density of sampling points, c_i is a value indicating a color of sampling points, and $\delta_{\text{sub}.i}$ is a value indicating a distance to the next sampling point.

[0061] After **S1002**, in **S1003**, the learning unit **304** updates the three-dimensional information so that a difference between the rendering value of the ray calculated in **S1002** and the value concerning the color of the ray included in the ray information becomes small for each piece of ray information included in the ray group of interest. Specifically, the learning unit **304** updates the three-dimensional information by updating a parameter of the function (learning model) indicating radiance fields so that the above difference becomes small. The calculation process of the difference between the rendering value of the ray and the value concerning the color of the ray and the updating process of the learning model described above correspond to an error calculation process and an error propagation process in deep learning. Incidentally, the difference between the rendering value of the ray and the value concerning the color of the ray is defined by, for example, a squared Euclidean distance of a color value specified by R (red), G (green), and B (blue).

[0062] Next, in **S1004**, the learning unit **304** judges whether all the ray groups have been selected in **S1001**. In a case where the learning unit **304** judges in **S1004** that not all the ray groups have been selected, that is, there is an unselected ray group, the learning unit **304** returns to **S1001** and repeats the process from **S1001** to **S1004**. In a case where the learning unit **304** judges in **S1004** that all the ray groups have been selected, the learning unit **304** finishes the process of the flowchart shown in FIG. 10, namely the process of **S404** shown in FIG. 4.

<Advantageous Result Produced by the Image Processing Apparatus According to the First Embodiment>

[0063] As described above, the image processing apparatus **102** learns radiance fields of a space including an object based on data on a plurality of captured images obtained by capturing the object with a plurality of image capturing apparatuses from various directions. In particular, in the present embodiment, the image processing apparatus **102** is configured to learn radiance fields, which are three-dimensional information on the image capturing region, using ray groups selected so that the density of rays is high in a space including the high-frequency pattern based on the high-frequency pattern region extracted from the captured images. According to the image processing apparatus **102** thus configured, three-dimensional information which enables generation of a high-quality virtual viewpoint image can be estimated while reducing the amount of computations for estimating the three-dimensional information. In particular, according to the image processing apparatus **102**, three-dimensional information on a space corresponding to an object including the high-frequency pattern can be estimated with high accuracy. As a result, the image quality of a virtual viewpoint image generated based on the estimated three-dimensional information can be improved.

Modified Example 1 of First Embodiment

[0064] The extraction unit **302** according to the first embodiment executes in **S402** the filtering processing using the Laplacian filter, the closing processing, and the opening processing and thereby extracts the high-frequency pattern region from the captured image. However, the method of extracting the high-frequency pattern region in the extraction unit **302** is not limited to the above method. For example, the extraction unit **302** may use any other differential filter such as a Sobel filter instead of the Laplacian filter to extract edges from the captured image. Further, for example, instead of the closing processing and the opening processing, the extraction unit **302** may specify a local region having a large proportion of an edge region and extract the specified local region as a high-frequency pattern region. Further, for example, an image including a representation corresponding to a specific object defined as a high-frequency pattern may be prepared as a template image in advance and the extraction unit **302** may extract a high-frequency pattern region by template matching using the template image. Further, for example, the extraction unit **302** may use a detector which detects from an input image a representation corresponding to a specific object predefined as a high-frequency pattern to extract the result of detection by the detector as a high-frequency pattern region.

[0065] Further, the ray group generation unit **303** according to the first embodiment generates in **S702** ray groups including only the ray information corresponding to the high-frequency pattern region, but the method of generating ray groups in the ray group generation unit **303** is not limited to the above method. For example, the ray group generation unit **303** may generate ray groups including both of the ray information corresponding to the high-frequency pattern region and the residual ray information so that the number of pieces of ray information corresponding to the high-frequency pattern region is equal to or greater than a predetermined number.

[0066] Specifically, for example, the number of pieces of ray information corresponding to the high-frequency pattern region and the number of pieces of residual ray information, to be included in one ray group are set in advance. Based on these preset numbers, the ray group generation unit **303** randomly selects ray information from both of all the pieces of ray information corresponding to the high-frequency pattern region and all the pieces of residual ray information and thereby generates ray groups. For example, the number of pieces of ray information corresponding to the high-frequency pattern region to be included in one ray group is set so that a ratio of the number of pieces of ray information corresponding to the high-frequency pattern region to the number of all the pieces of ray information to be included in one ray group is equal to or greater than a predetermined ratio. Here, the predetermined ratio is, for example, a ratio of the number of pixels included in the high-frequency pattern region to the number of all the pixels of the captured image.

[0067] FIG. **8B** shows an example of ray groups **813** to **816** each including the ray information corresponding to the high-frequency pattern region and the residual ray information generated through the above process by the ray group generation unit **303**. Ray groups **817** to **824** shown in FIG. **8B** are an example of ray groups including only the residual ray information generated by the ray group generation unit **303** executing the process of **S703** after generating the ray groups **813** to **816**.

[0068] Further, the ray group generation unit **303** according to the first embodiment generates in **S702** and **S703** a plurality of ray groups so that the same piece of ray information is not included in more than one ray group. However, the ray group generation unit **303** may generate a plurality of ray groups while allowing the same piece of ray information to be included in more than one ray group. In this case, the total number of ray groups may be equal to that in the case where the same piece of ray information is not included in more than one ray group.

[0069] Further, the ray group generation unit **303** according to the first embodiment calculates in **S701** ray information corresponding to each pixel of the captured image. However, the ray group generation unit **303** may store in the RAM **202** or the like the ray information calculated in the first execution of **S701** and obtain the ray information stored in the RAM **202** or the like in the second

execution of **S701** onward.

[0070] Further, the three-dimensional information according to the first embodiment is described as a radiance field, which is a function (learning model) expressed by a multilayer perceptron, but the method of expressing the radiance field is not limited to this. For example, the radiance field may be expressed by using a plurality of multilayer perceptrons, a sparse three-dimensional grid including spherical surface harmonics, or a tensor.

[0071] Further, the learning unit **304** according to the first embodiment is described as learning the learning model indicating the radiance field, but the learning unit **304** may learn a learning model indicating three-dimensional information which can be learned based on ray groups. For example, although the radiance field indicates a color and density according to a position and orientation, the three-dimensional information is not limited to this. Specifically, for example, a color corresponding to a position in a space in the three-dimensional information may be a color having isotropy not dependent on a direction. Further, for example, the three-dimensional information may be a density field indicating a density according to a position, a field expressed by a bidirectional reflectance distribution function indicating distribution characteristics of reflected light with respect to incident light, or a field expressing light visibility of environmental light. Further, the three-dimensional information may be a field indicating a color and density corresponding to a position, direction, and time. In this case, ray groups for use in learning of the three-dimensional information are generated based on captured images as moving images including time series frames.

Second Embodiment

[0072] The image processing apparatus **102** according to the second embodiment will be described with reference to FIGS. **2** to **4** and **11** to **16**. The image processing apparatus **102** according to the present embodiment has the hardware configuration and functional configuration illustrated in the block diagrams of FIG. **2** or **3** like the image processing apparatus **102** according to the first embodiment. The image processing apparatus **102** according to the present embodiment executes the process of the flowchart illustrated in FIG. **4** like the image processing apparatus **102** according to the first embodiment. However, in the present embodiment, the processes in the extraction unit **302** and the ray group generation unit **303** are different from those in the first embodiment. That is, the extraction process of the high-frequency pattern region in **S402** and the generation process of ray groups in **S403** according to the present embodiment are different from those in the first embodiment.

[0073] Specifically, the extraction unit **302** according to the first embodiment extracts the high-frequency pattern region from the captured image using a differential filter such as a Laplacian filter in the extraction process of the high-frequency pattern region in **S402**. In contrast, the extraction unit **302** according to the present embodiment extracts, as a high-frequency pattern region, an image region corresponding to an approximate shape of each object including the high-frequency pattern from the captured image based on the approximate shape of the object. Further, the ray group generation unit **303** according to the present embodiment generates ray groups for each high-frequency pattern region extracted by the extraction unit **302**. The following description mainly describes the extraction process of the high-frequency pattern region in the extraction unit **302** and the generation process of ray groups in the ray group generation unit **303** in the present embodiment, which are different from the processes in the first embodiment. It should be noted that a configuration or processing step which performs the same process as that in the first embodiment is denoted by the same reference numeral and its description is omitted.

<Extraction Process of High-Frequency Pattern Region>

[0074] FIG. **11** is a flowchart showing an example of the flow of the extraction process of the high-frequency pattern region in the extraction unit **302** according to the second embodiment and showing an example of the detailed processing flow in **S402** shown in FIG. **4**. In **S402** according to the second embodiment, an image region corresponding to an approximate shape including the high-frequency pattern is extracted as a high-frequency pattern region from the captured image

obtained in **S401** based on the approximate shape of the object. The process of this flowchart is executed after the process of **S401** shown in FIG. 4.

[0075] After **S401**, in **S1101**, the extraction unit **302** first extracts a region with high edge density from each captured image obtained in **S401** like the first embodiment. FIGS. **12A** to **12D** are diagrams for explaining the extraction process of the high-frequency pattern region in the extraction unit **302** according to the second embodiment. FIG. **12A** shows an example of a captured image **1200** obtained through image capturing by an image capturing apparatus **101**. It is assumed that the captured image **1200** shown in FIG. **12A** includes representations **1201** to **1203** of three objects existing in the image capturing region **106** as an example. It is also assumed that the high-frequency pattern is included in the object corresponding to the representation **1201** (hereinafter referred to as “object A”) and the object corresponding to the representation **1203** (hereinafter referred to as “object B”) among the representations **1201** to **1203** included in the captured image **1200**.

[0076] FIG. **12B** shows an example of an image **1210** indicating the region with high edge density extracted in **S1101**. The image **1210** shown in FIG. **12B** is shown as a binary image in which a pixel value of the region with high edge density is 1 and a pixel value of a region with low edge density is 0 as an example. In the above example, a region **1211** corresponding to the representation **1201** of the object A and a region **1213** corresponding to the representation **1203** of the object B are extracted in **S1101** as regions with high edge density. FIGS. **12C** and **12D** will be described later in detail.

[0077] After **S1101**, in **S1102**, the extraction unit **302** obtains an approximate shape of the object based on the captured images and image capturing parameters obtained in **S401**. In the present embodiment, the extraction unit **302** is assumed to obtain an approximate shape of the object expressed as a set of voxels by a visual hull method. In this case, for example, the extraction unit **302** first obtains a silhouette image of the object for each captured image obtained in **S401** based on a difference between the captured image and a background image obtained by capturing only a background with no object. For example, the background image may be captured in advance and obtained by the extraction unit **302** reading data on the background image prestored in the storage device **204** or the like. Since the method of obtaining the silhouette image of the object is well known, its detailed description is omitted.

[0078] Next, the extraction unit **302** projects each voxel included in the set of voxels corresponding to the image capturing region on the silhouette image based on the image capturing parameter obtained in **S401**. The extraction unit **302** then obtains, as an approximate shape of the object, a set of voxels projected on a silhouette region of the object for all the silhouette images. FIG. **13** is a diagram showing an example of an approximate shape of the object **1301** obtained by the visual hull method. A rectangle surrounded by thin solid lines shows an actual outline **1302** of the object **1301** and a polygon surrounded by thick solid lines shows an outline **1303** of the approximate shape of the object **1301**. Since the method of obtaining an approximate shape of an object such as a visual hull method based on a silhouette image of the object is well known, its detailed description is omitted.

[0079] After **S1102**, in **S1103**, the extraction unit **302** extracts an approximate shape of an object including the high-frequency pattern from approximate shapes of a plurality of objects obtained in **S1102**. Specifically, for example, the extraction unit **302** first projects each voxel corresponding to the surface of the approximate shape on each captured image using the corresponding image capturing parameter. Next, the extraction unit **302** extracts, as an approximate shape of an object including the high-frequency pattern, such an approximate shape that the number of voxels projected on a region with high edge density in the captured image is equal to or greater than a predetermined value while the approximate shape of the object is not occluded by any of the approximate shapes of the other objects.

[0080] Next, in **S1104**, the extraction unit **302** extracts, as a high-frequency pattern region, an

image region in the captured image corresponding to the approximate shape of the object including the high-frequency pattern extracted in **S1103**. Specifically, the extraction unit **302** extracts, as a high-frequency pattern region, such a set of pixels that rays corresponding to the pixels intersect with the approximate shape of the object including the high-frequency pattern without being occluded by any of the approximate shapes of the other objects. After **S1104**, the extraction unit **302** finishes the process of the flowchart shown in FIG. **11**, namely the process of **S402** according to the present embodiment.

[0081] FIG. **12C** shows an example of the relationship between approximate shapes **1224** and **1225** of objects including the high-frequency pattern extracted in **S1103** and high-frequency pattern regions **1226** and **1227** extracted in **S1104**. In FIG. **12C**, objects **1221** to **1223** are objects existing in the image capturing region **106** and the approximate shapes **1224** and **1225** are approximate shapes of the objects **1221** and **1223** including the high-frequency pattern, respectively. In FIG. **12C**, the high-frequency pattern regions **1226** and **1227** are high-frequency pattern regions corresponding to the approximate shapes **1224** and **1225** of the objects including the high-frequency pattern, respectively. The objects **1221** and **1223** correspond to the above objects A and B, respectively.

[0082] FIG. **12D** shows an example of an image **1230** showing the high-frequency pattern regions **1226** and **1227** extracted in **S1104**. In the image **1230** shown in FIG. **12D**, the high-frequency pattern region **1226** corresponds to the approximate shape **1224** of the object **1221** including the high-frequency pattern. The high-frequency pattern region **1227** corresponds to the approximate shape **1225** of the object **1223** including the high-frequency pattern. In the following description, the high-frequency pattern region **1226** is referred to as “high-frequency pattern region A” and the high-frequency pattern region **1227** is referred to as “high-frequency pattern region B.”

<Generation Process of Ray Groups>

[0083] FIG. **14** is a flowchart showing an example of the flow of the generation process of ray groups in the ray group generation unit **303** according to the second embodiment and showing an example of the detailed processing flow in **S403** shown in FIG. **4**. In **S403** according to the present embodiment, ray groups are generated for each high-frequency pattern region extracted in **S402** according to the present embodiment. The process of this flowchart is executed after the process of **S402** according to the present embodiment.

[0084] After **S402** according to the present embodiment, the ray group generation unit **303** first executes the same process as **S701** shown in FIG. **7** and obtains ray information corresponding to each pixel of the captured images based on the captured images and image capturing parameters obtained in **S401**. Next, in **S1401**, the ray group generation unit **303** selects an arbitrary high-frequency pattern region from one or more high-frequency pattern regions extracted in **S402**. Specifically, the ray group generation unit **303** randomly selects an arbitrary unselected high-frequency pattern region of the one or more high-frequency pattern regions. The high-frequency pattern region selected in **S1401** is hereinafter referred to as “region of interest.”

[0085] Next, in **S1402**, the ray group generation unit **303** generates a plurality of ray groups based on ray information corresponding to image coordinates of each pixel included in the region of interest (hereinafter referred to as “ray information corresponding to the region of interest”). Like the first embodiment, one ray group includes a plurality of pieces of ray information. Further, like the first embodiment, it is assumed in the present embodiment that a plurality of ray groups are generated so that the same piece of ray information corresponding to the region of interest is not included in more than one ray group. Specifically, the ray group generation unit **303** defines a predetermined number of pieces of ray information randomly selected from the ray information corresponding to the region of interest as one ray group. More specifically, the ray group generation unit **303** generates a plurality of ray groups so that the same ray information in ray information corresponding to the region of interest is not selected multiple times. The ray group generation unit **303** repeats the above process until the number of unselected pieces of ray

information corresponding to the region of interest is less than a predetermined number. In this manner, a plurality of ray groups are generated so that the same piece of ray information corresponding to the region of interest is not included in more than one ray group. Incidentally, unselected pieces of ray information corresponding to the region of interest may be used as residual ray groups in the process of **S1405** to be described later.

[0086] FIGS. **15A** and **15B** are diagrams for explaining examples of ray groups generated by the ray group generation unit **303**. Specifically, FIG. **15A** is a diagram for explaining an example of ray groups according to the second embodiment and FIG. **15B** is a diagram for explaining an example of ray groups according to modified example 1 of the second embodiment to be described later. In FIGS. **15A** and **15B**, rectangles **1541** filled in dark gray represent ray information corresponding to the high-frequency pattern region A and rectangles **1542** filled in light gray represent ray information corresponding to the high-frequency pattern region B. Rectangles **1543** filled in white represent ray information (residual ray information) corresponding to image coordinates of each pixel included in a region (residual region) other than the high-frequency pattern regions A and B.

[0087] A set of ray information **1551** is a set of all pieces of ray information corresponding to the high-frequency pattern region A. Specifically, for example, the set of ray information **1551** indicates a set of pieces of ray information which are selectable by the ray group generation unit **303** for generating a plurality of ray groups in **S1402** in a case where the region of interest is the high-frequency pattern region A. A set of ray information **1552** is a set of all pieces of ray information corresponding to the high-frequency pattern region B. Specifically, for example, the set of ray information **1552** indicates a set of pieces of ray information which are selectable by the ray group generation unit **303** for generating a plurality of ray groups in **S1402** in a case where the region of interest is the high-frequency pattern region B. A set of ray information **1553** indicates a set of all the residual ray information. A set of ray information **1550** indicates a set of pieces of ray information corresponding to image coordinates of all the pixels included in the captured image, including all the pieces of ray information corresponding to the high-frequency pattern regions A and B and all the residual ray information.

[0088] In a case where the region of interest is the high-frequency pattern region A, the ray group generation unit **303** executes the process of **S1402** and thereby generates a plurality of ray groups **1501** to **1503** illustrated in FIG. **15A**, each including only the ray information corresponding to the high-frequency pattern region A. In a case where the region of interest is the high-frequency pattern region B, the ray group generation unit **303** executes the process of **S1402** and thereby generates a plurality of ray groups **1504** to **1506** shown in FIG. **15A**, each including only the ray information corresponding to the high-frequency pattern region B. A ray corresponding to each piece of ray information included in the ray groups **1501** to **1506** thus generated intersects with the object A or B including the high-frequency pattern.

[0089] FIGS. **16A** and **16B** are diagrams for explaining an example of a ray **1605** corresponding to each piece of ray information included in the ray groups **1501** to **1503** shown in FIG. **15A**. Specifically, FIG. **16A** is a diagram of the image capturing region **106** viewed vertically from above and shows an example of the ray **1605** corresponding to each piece of ray information included in the ray groups **1501** to **1503** shown in FIG. **15A**. In FIG. **15A**, the objects **1601** and **1603** are objects including the high-frequency pattern and the object **1602** is an object not including the high-frequency pattern. FIG. **16B** shows an example of a captured image **1610** obtained through image capturing by the image capturing apparatus **101c**. The captured image **1610** shown in FIG. **16B** includes representations **1611** to **1613** of the three objects **1601** to **1603** existing in the image capturing region **106**.

[0090] As described above, the ray **1605** corresponding to each piece of ray information included in the ray groups **1501** to **1503** intersects with the object **1601** including the high-frequency pattern. Further, although not shown in FIG. **16A**, a ray corresponding to each piece of ray information included in the ray groups **1504** to **1506** intersects with the object **1603** including the high-

frequency pattern. Accordingly, even in a case where there are a plurality of objects including the high-frequency pattern, the density of rays passing through the space including the high-frequency pattern is higher than the density of rays passing through the other space. Therefore, learning can be performed using high-density ray groups for the space corresponding to the objects **1601** and **1603** including the high-frequency pattern. As a result, the three-dimensional information (radiance field) of the space corresponding to the objects including the high-frequency pattern can be estimated with high accuracy.

[0091] After **S1402**, in **S1403**, the ray group generation unit **303** judges whether all the high-frequency pattern regions extracted in **S402** according to the present embodiment have been selected in **S1402**. In a case where the ray group generation unit **303** judges in **S1403** that not all the high-frequency pattern regions have been selected, that is, there is an unselected high-frequency pattern region, the ray group generation unit **303** returns to **S1401** and repeats the process from **S1401** to **S1403**. In a case where the ray group generation unit **303** judges in **S1403** that all the high-frequency pattern regions have been selected, the ray group generation unit **303** executes the process of **S1404**. In **S1404**, the ray group generation unit **303** executes the same process as **S703** shown in FIG. 7 and thereby generates a plurality of ray groups based on the residual ray information. Incidentally, ray groups **1507** to **1512** shown in FIG. 15A are an example of ray groups including only the residual ray information generated by the ray group generation unit **303** executing the process of **S1404**.

[0092] After **S1404**, the ray group generation unit **303** finishes the process of flowchart shown in FIG. 14, namely the process of **S403** according to the present embodiment. That is, the ray group generation unit **303** according to the present embodiment executes the process of **S403** and thereby generates a plurality of ray groups **1501** to **1512** from the set of ray information **1550** corresponding to image coordinates of all the pixels included in the captured image as shown in FIG. 15A.

Modified Example 1 of Second Embodiment

[0093] The ray group generation unit **303** according to the second embodiment generates in **S1402** ray groups including only the ray information corresponding to the high-frequency pattern region set as the region of interest. However, the method of generating ray groups in the ray group generation unit **303** is not limited to the above method. For example, the ray group generation unit **303** may generate ray groups including both the ray information corresponding to the region of interest and the residual ray information so that the number of pieces of ray information corresponding to the region of interest is equal to or greater than a predetermined number.

[0094] Specifically, for example, the number of pieces of ray information corresponding to the region of interest and the number of pieces of residual ray information to be included in one ray group are set in advance. Based on these preset numbers, the ray group generation unit **303** randomly selects ray information from both of all the pieces of ray information corresponding to the region of interest and all the pieces of residual ray information and thereby generates ray groups. For example, the number of pieces of ray information corresponding to the region of interest to be included in one ray group is set so that a ratio of the number of pieces of ray information corresponding to the region of interest to the number of all the pieces of ray information to be included in one ray group is equal to or greater than a predetermined ratio. The predetermined ratio is, for example, a ratio of the number of pixels included in the region of interest to the number of all the pixels of the captured image.

[0095] FIG. 15B shows an example of ray groups **1513** to **1516** including the ray information corresponding to the high-frequency pattern region A and the residual ray information generated through the above process by the ray group generation unit **303**. FIG. 15B also shows an example of ray groups **1517** to **1520** including the ray information corresponding to the high-frequency pattern region B and the residual ray information generated through the above process by the ray group generation unit **303**. Ray groups **1521** to **1524** shown in FIG. 15B are an example of ray

groups including only the residual ray information generated by the ray group generation unit **303** executing the process of **S1404** after generating the ray groups **1513** to **1520**.

[0096] Further, the ray group generation unit **303** according to the second embodiment generates in **S1402** and **S1404** a plurality of ray groups so that the same piece of ray information is not included in more than one ray group. However, the ray group generation unit **303** may generate a plurality of ray groups while allowing the same piece of ray information to be included in more than one ray group. In this case, the total number of ray groups may be equal to that in the case where the same piece of ray information is not included in more than one ray group.

[0097] Further, the extraction unit **302** according to the second embodiment obtains in **S1102** the approximate shape of the object by the visual hull method, but the method of obtaining the approximate shape of the object is not limited to this. For example, the extraction unit **302** may obtain the approximate shape of the object based on distance information obtained by stereo matching or the like using the captured images obtained by the image obtaining unit **301** or distance images obtained by a depth camera. Further, information indicating the approximate shape of the object may be generated in advance and stored in the storage device **204** or the like and the extraction unit **302** may obtain the approximate shape of the object by reading the information.

[0098] Further, instead of the approximate shape of the object, the extraction unit **302** may specify a local space including the object among a plurality of local spaces obtained by dividing the image capturing region **106** and obtain a set of specified local spaces including the object as the approximate shape of the object. In this case, for example, the extraction unit **302** may obtain a local space including the approximate shape of the object as a local space including the object. Further, for example, the extraction unit **302** may obtain a local space including the object by the following method: specifically, the extraction unit **302** first extracts a feature point from each captured image obtained by the image obtaining unit **301**, associates the extracted feature point of the captured images with each other, projects the feature points on a space, and then obtains a local space having a high density of projected feature points as a local space including the object.

<Advantageous Result Produced by the Image Processing Apparatus According to the Second Embodiment>

[0099] As described above, the image processing apparatus **102** according to the second embodiment extracts, as a high-frequency pattern region, an image region corresponding to an approximate shape of an object from the captured image for each object including the high-frequency pattern based on the approximate shape of the object. The image processing apparatus **102** also generates ray groups for each high-frequency pattern region. According to the image processing apparatus **102** thus configured, three-dimensional information which enables generation of a high-quality virtual viewpoint image can be estimated while reducing the amount of computations for estimating the three-dimensional information. In particular, according to the image processing apparatus **102**, three-dimensional information on a space corresponding to a plurality of objects including the high-frequency pattern can be estimated with high accuracy. As a result, the image quality of a virtual viewpoint image generated based on the estimated three-dimensional information can be improved.

Other Embodiments

[0100] Some embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer-executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer-executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling

the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer-executable instructions. The computer-executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory apparatus, a memory card, and the like.

[0101] According to the technique of this disclosure, three-dimensional information which enables generation of a high-quality virtual viewpoint image can be estimated while reducing the amount of computations for estimating the three-dimensional information.

[0102] While the present disclosure has described exemplary embodiments, it is to be understood that some embodiments of the disclosure are not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0103] This application claims priority to Japanese Patent Application No. 2024-032877, filed on Mar. 5, 2024, which is hereby incorporated by reference wherein in its entirety.

Claims

1. An image processing apparatus comprising: one or more hardware processors; and one or more memories storing one or more programs configured to be executed by the one or more hardware processors, the one or more programs including instructions for: obtaining data on a captured image; extracting a high-frequency pattern region from the captured image; generating a plurality of ray groups based on rays corresponding to each of a plurality of pixels included in the captured image; generating the plurality of ray groups including one or more high-frequency pattern region ray groups which are ray groups where a density of rays corresponding to pixels included in the high-frequency pattern region is higher than a density of rays corresponding to pixels included in a region other than the high-frequency pattern region; and learning three-dimensional information on a space using the plurality of ray groups.
2. The image processing apparatus according to claim 1, wherein the one or more programs further include instructions for: generating the high-frequency pattern region ray groups by randomly selecting a plurality of rays from the rays corresponding to the each of a plurality of pixels included in the high-frequency pattern region.
3. The image processing apparatus according to claim 1, wherein the one or more programs further include instructions for: generating the high-frequency pattern region ray groups so that a ratio of a total number of rays corresponding to the pixels included in the high-frequency pattern region to a total number of rays included in the high-frequency pattern region ray groups is equal to or greater than a predetermined ratio.
4. The image processing apparatus according to claim 3, wherein the predetermined ratio is equal to or greater than a ratio of a total number of the pixels included in the high-frequency pattern region to a total number of the pixels included in the captured image.
5. The image processing apparatus according to claim 1, wherein the one or more programs further include instructions for: generating the high-frequency pattern region ray groups so that a predetermined number or more of rays corresponding to the pixels included in the high-frequency pattern region are included.
6. The image processing apparatus according to claim 5, wherein the predetermined number is such a number that a ratio of the predetermined number to a total number of rays included in the high-frequency pattern region ray groups is equal to or greater than a ratio of a total number of the pixels

included in the high-frequency pattern region to a total number of the pixels included in the captured image.

7. The image processing apparatus according to claim 1, wherein the one or more programs further include instructions for: generating the high-frequency pattern region ray groups including only rays corresponding to the pixels included in the high-frequency pattern region.

8. The image processing apparatus according to claim 1, wherein the one or more programs further include instructions for: extracting the high-frequency pattern region based on a local space corresponding to an object including a high-frequency pattern.

9. The image processing apparatus according to claim 8, wherein the one or more programs further include instructions for: extracting the high-frequency pattern region for each of a plurality of objects including the high-frequency pattern based on the local space corresponding to the object.

10. The image processing apparatus according to claim 8, wherein the local space is an approximate shape of the object included as a representation in the captured image.

11. The image processing apparatus according to claim 1, wherein the high-frequency pattern region is an image region of the captured image where an edge density is higher than a predetermined density.

12. The image processing apparatus according to claim 1, wherein the one or more programs further include instructions for: generating the high-frequency pattern region ray groups by selecting a plurality of rays so that each of the rays corresponding to the pixels included in the high-frequency pattern region is not selected more than once.

13. An image processing method comprising the steps of: obtaining data on a captured image; extracting a high-frequency pattern region from the captured image; generating a plurality of ray groups based on rays corresponding to each of a plurality of pixels included in the captured image; generating the plurality of ray groups including one or more high-frequency pattern region ray groups which are ray groups where a density of rays corresponding to pixels included in the high-frequency pattern region is higher than a density of rays corresponding to pixels included in a region other than the high-frequency pattern region; and learning three-dimensional information on a space using the plurality of ray groups.

14. A non-transitory computer readable storage medium storing a program for causing a computer to perform a control method of an image processing apparatus, the control method comprising the steps of: obtaining data on a captured image; extracting a high-frequency pattern region from the captured image; generating a plurality of ray groups based on rays corresponding to each of a plurality of pixels included in the captured image; generating the plurality of ray groups including one or more high-frequency pattern region ray groups which are ray groups where a density of rays corresponding to pixels included in the high-frequency pattern region is higher than a density of rays corresponding to pixels included in a region other than the high-frequency pattern region; and learning three-dimensional information on a space using the plurality of ray groups.
